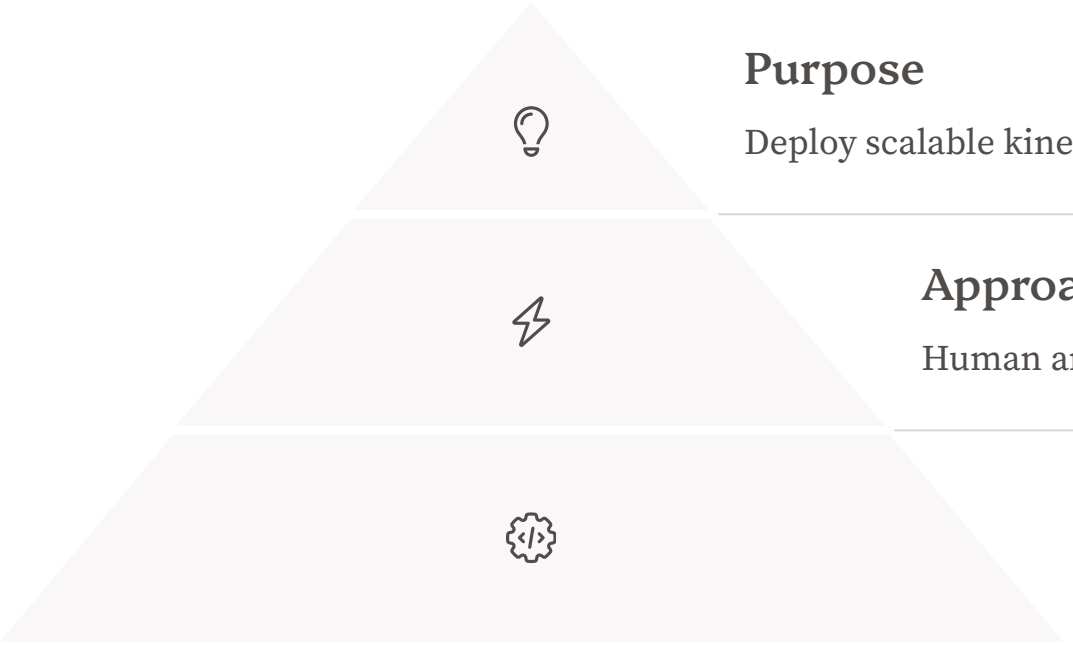
Vedavathi Kinetic Energy & Smart Energy Innovations Strategy

Welcome to Vedavathi's comprehensive strategy for deploying innovative kinetic and smart micro-energy solutions across our operations. This presentation outlines our plan to harness decentralized, human and livestock-powered energy harvesting systems that deliver localized ROI and operational efficiency.

Join us as we explore how these grassroots technologies will transform our energy landscape while supporting our commitment to sustainability and community empowerment.

R **by Raja Suri**

Our Purpose & Innovation Domains



Purpose

Deploy scalable kinetic and smart micro-energy innovations

Approach

Human and livestock-powered energy harvesting systems

Focus

Decentralized, grassroots technologies with localized ROI

Our strategy focuses on five key innovation domains: cattle-powered kinetic systems, cycle-powered charging stations, kinetic floor tiles, water flow turbines, and agricultural recovery units. These technologies will be integrated across farms, walking trails, livestock areas, corporate offices, schools, riverbanks, and irrigation infrastructure.

Technology Stack & Implementation

Piezoelectric Floor Units

Convert footsteps into electrical energy

Rotary Mechanical-to-Electrical Drives

Harvest energy from rotating machinery

Kinetic-Dynamo Hybrid Chains

Capture energy from cycles and carts

Smart Conversion Hubs + Li-ion Storage

Store and distribute harvested energy efficiently

AI-Enabled Usage Prediction

Optimize load routing and distribution

Our implementation approach uses advanced technologies that convert various forms of movement into usable electricity. This comprehensive stack ensures we capture energy from multiple sources while optimizing storage and distribution.

Bellary: Cycle & Cattle Power Hub

Technology Deployment

- High-density cycle-to-battery stations
- Rotary cattle walk pads
- Borewell-integrated turbines

Expected Output

7+ MW of clean energy generation

Primary applications include smart grid load management, dairy refrigeration, and powering internal EV fleets

Key Vendors

- CleanMotion
- eMotion Energies
- Kinetic Energy Group

Bellary will serve as our flagship location for integrated kinetic energy solutions, combining livestock movement with human-powered technologies to create a multi-source smart grid that generates over 5 MW of power.

Kurnool: Cycle Infrastructure & Tree Robotics

Kinetic Cycling Farms

Large-scale facilities with multiple cycle-power stations generating significant clean energy

Robotic Tree Maintenance

Worker paths under maintenance corridors capturing kinetic energy from daily activities

Canal Turbines

Water-powered generators installed in irrigation infrastructure

At Kurnool, we're integrating cycle infrastructure with our tree robotics program to maximize energy capture. With an expected output of 6 MW, this power will support automation nodes, field sensors, and smart pumping systems. Vendors include Lucid Energy, EnGoPlanet, and Urban Motion Tech.

Gadwal: Dairy Kinetics & Regional Cycling

Cattle Rotary Systems

Medium-scale energy harvesting from livestock movement

Cycle Hubs

Strategically placed near dairy zones for efficient energy capture

Vendors

Sanyo SmartKinetics, PowerSpout, EcoMotion Solutions

Applications

Shed lighting, agricultural cooling, remote lighting

Our Gadwal site will focus on dairy kinetics and regional cycling, generating approximately 4 MW of power. This energy will primarily support dairy operations, providing sustainable power for refrigeration, lighting, and small motors across the facility.

Hyderabad: Urban Demonstration & Micro Energy

Urban Demonstration

Kinetic tiles and floor systems at entry-exit points that convert footsteps into electricity

Cycle-charging kiosks for visitors to experience energy generation firsthand

Expected output: 50-100 kW

Applications

Campus lighting systems powered by harvested energy

Digital signage in lobbies and visitor areas

Educational displays demonstrating energy conversion

Vendor Partners

Pavegen (UK) - Leaders in kinetic flooring technology

Powerleap - Innovative energy harvesting solutions

Urban Green Tech - Integrated urban energy systems

Our Hyderabad locations will showcase urban applications of kinetic energy technology, serving both practical power generation and educational purposes. These installations will demonstrate our commitment to innovation while providing backup power for essential systems.

Kurnool: Agri Mobility & Canal Integration

Irrigation Canal Turbines

Small-scale hydro generators installed in existing canal infrastructure to capture energy from water flow without disrupting irrigation operations

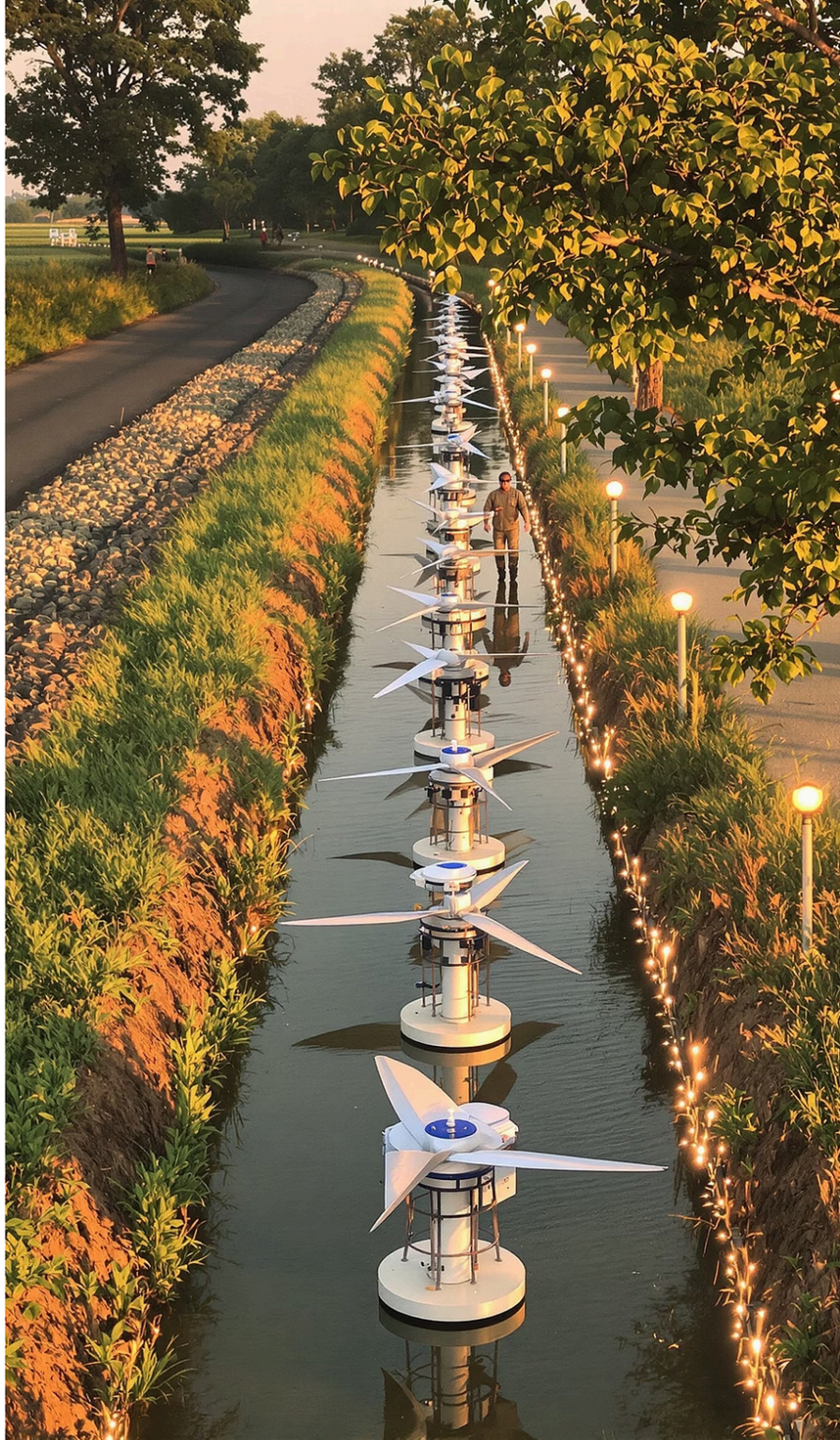
Kinetic Pathways

Energy-harvesting walkways installed along common routes used by tree workers, converting footsteps into electricity to power nearby systems

Smart Distribution

Integrated network connecting harvested energy to pump controls, telemetry stations, and lighting systems across the agricultural operation

At our Kurnool agricultural sites, we'll generate 1-2 MW by combining irrigation canal turbines with kinetic pathways. This energy will power critical pump controls, telemetry stations, and lighting systems. Technology partners include WindStream Technologies and Lucid Energy.



Return on Investment Analysis

Location	System Type	CapEx (Cr)	Monthly Savings (Cr)	ROI Period
Hyderabad	Cycle + Pedestrian	50	7	7.1 months
Kurnool	Agri-canal kinetic	250	35	7.1 months
Gadwal	Livestock energy harvest	180	30	6.0 months
Bellary	Integrated smart grid	400	60	6.7 months

Our financial analysis demonstrates the compelling economics of kinetic energy systems. Across all locations, we expect ROI periods between 6-7.1 months, with the fastest returns coming from our livestock energy harvesting systems at Gadwal.

These impressive payback periods are driven by substantial monthly savings and relatively modest capital expenditures compared to traditional energy infrastructure.

Implementation Workflow

Energy Mapping

Identify movement paths and water flows

Feasibility Study

Deploy sensors to measure energy potential

Partner Collaboration

Work with vendors on custom prototypes

Pilot Installations

Test systems at each site type

Our implementation begins with comprehensive energy mapping to identify optimal placement for kinetic harvesting systems. After feasibility studies with sensor placement, we'll collaborate with partners on custom prototypes before installing pilots at each site type.

This systematic approach ensures we maximize energy capture while minimizing disruption to existing operations.

Technology Integration & Monitoring

Load Routing Development

Create intelligent distribution systems that direct harvested energy to where it's needed most

AI Prediction Model Deployment

Implement machine learning algorithms to forecast energy production and consumption patterns

Battery Unit Integration

Connect storage systems to stabilize supply and ensure consistent power availability

Monitoring Dashboard Setup

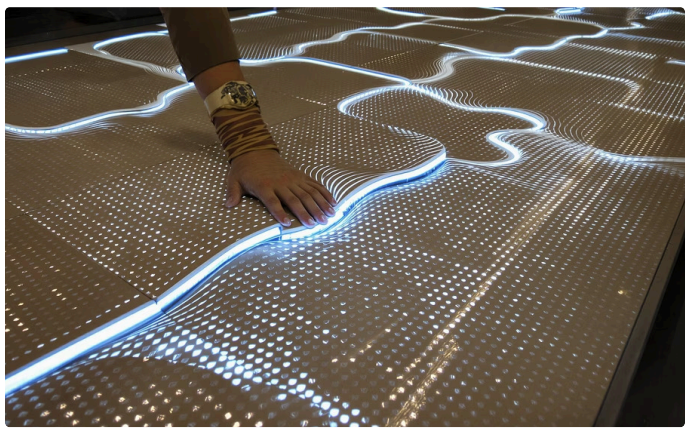
Deploy comprehensive visualization tools to track performance across all systems

ROI Benchmarking & Scaling

Analyze performance data to validate ROI and identify scaling opportunities

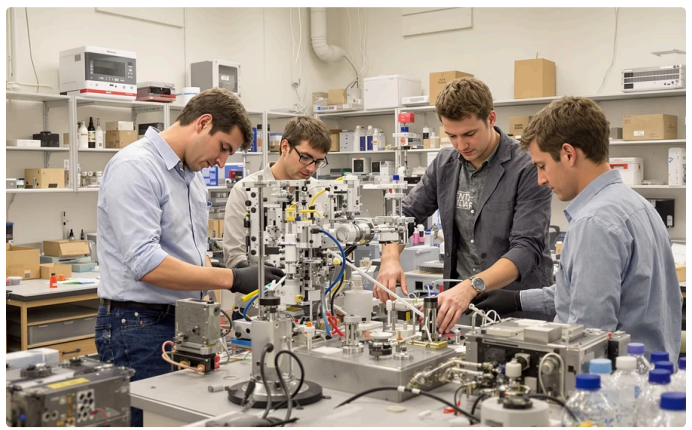
After installation, we'll focus on sophisticated integration with AI-powered load routing, battery storage, and comprehensive monitoring systems. This technical infrastructure will optimize performance while providing valuable data for continuous improvement.

Strategic Partnerships



Innovation Partners

Pavegen brings world-leading expertise in kinetic tile technology that converts footsteps into electricity. EnGoPlanet specializes in hybrid solar-kinetic systems, while eMotion provides cutting-edge mechanical energy harvesting solutions.



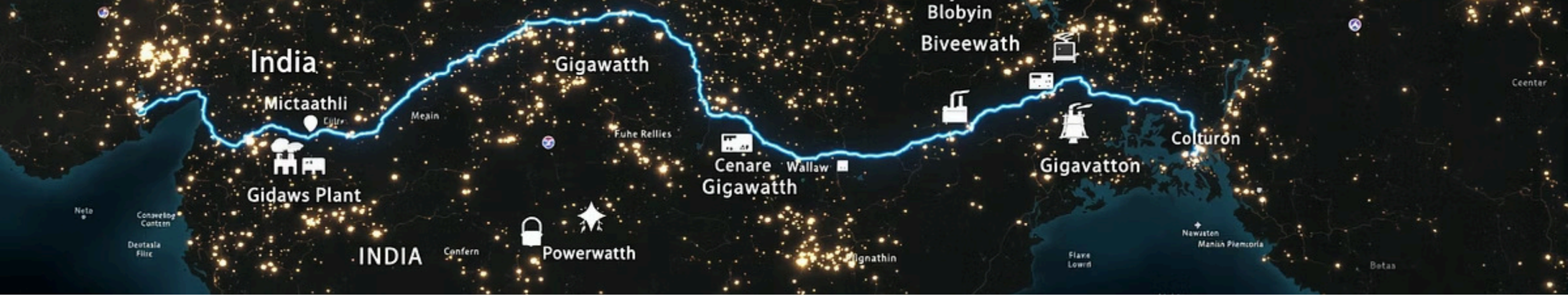
Engineering Support

We're collaborating with IIT incubation centers to access cutting-edge research and development capabilities. Local mechatronics startups provide agile innovation and customization for our specific needs.

Battery Partners

Strategic relationships with Amara Raja, Tesla (Powerwall Micro), and Livguard ensure we have optimal energy storage solutions that integrate seamlessly with our kinetic harvesting systems.

Strong partnerships are fundamental to our strategy, providing specialized expertise across the technology spectrum from energy capture to storage and distribution.



Output and Applications by Location

7+ MW

Bellary

Smart grid load, dairy refrigeration, internal EV fleets

6 MW

Kurnool

Automation nodes, field sensors, smart pumps

4 MW

Gadwal

Shed lighting, agri-cooling, remote lighting

100 kW

Hyderabad

Campus lighting, lobby signage, awareness tools

Each location has been strategically selected to maximize energy output based on available resources and activities. From Bellary's impressive 7+ MW to Hyderabad's demonstrative 100 kW, these systems are tailored to local needs and opportunities.

The applications vary by location, ensuring that harvested energy directly supports core operations while reducing dependence on conventional power sources.

Benefits Beyond Energy Production

Environmental Impact

Our kinetic energy systems will reduce Vedavathi's carbon footprint by capturing energy that would otherwise be lost, supporting our sustainability commitments while setting new industry standards.

Community Engagement

These technologies create opportunities for community participation in energy generation, fostering a culture of sustainability while demonstrating our commitment to innovative solutions.

Operational Resilience

Decentralized energy production enhances our resilience against grid failures and energy price fluctuations, ensuring critical operations can continue even during conventional power outages.

Educational Value

Our installations will serve as living laboratories for students, employees, and visitors to learn about innovative energy solutions, inspiring the next generation of sustainability leaders.

The value of our kinetic energy initiative extends far beyond kilowatt production, creating meaningful environmental, social, and operational benefits across our organization and communities.

Moving Forward: Next Steps

Detailed Site Assessment

Begin comprehensive energy mapping and detailed engineering studies at priority locations

Stakeholder Workshops

Conduct sessions with location managers and key personnel to refine implementation strategies

Pilot Deployment

Launch first installations at strategic locations to validate technology and ROI projections

Vendor Selection

Finalize partnerships with key technology providers and engineering support teams

Kinetic energy and smart energy innovations represent a frontier opportunity for Vedavathi to pioneer decentralized, socially participative, and cost-effective energy generation. These systems will complement our large-scale infrastructure while delivering utility savings, off-grid resilience, and powerful grassroots brand impact.

By moving forward with these initiatives, we position Vedavathi as a leader in sustainable innovation while creating tangible benefits for our operations, communities, and the environment.